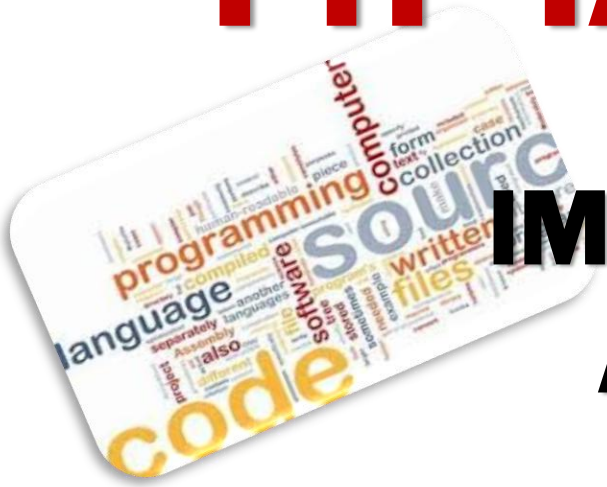
FYP TALK SEMINAR



IMPLEMENTATION AND TESTING



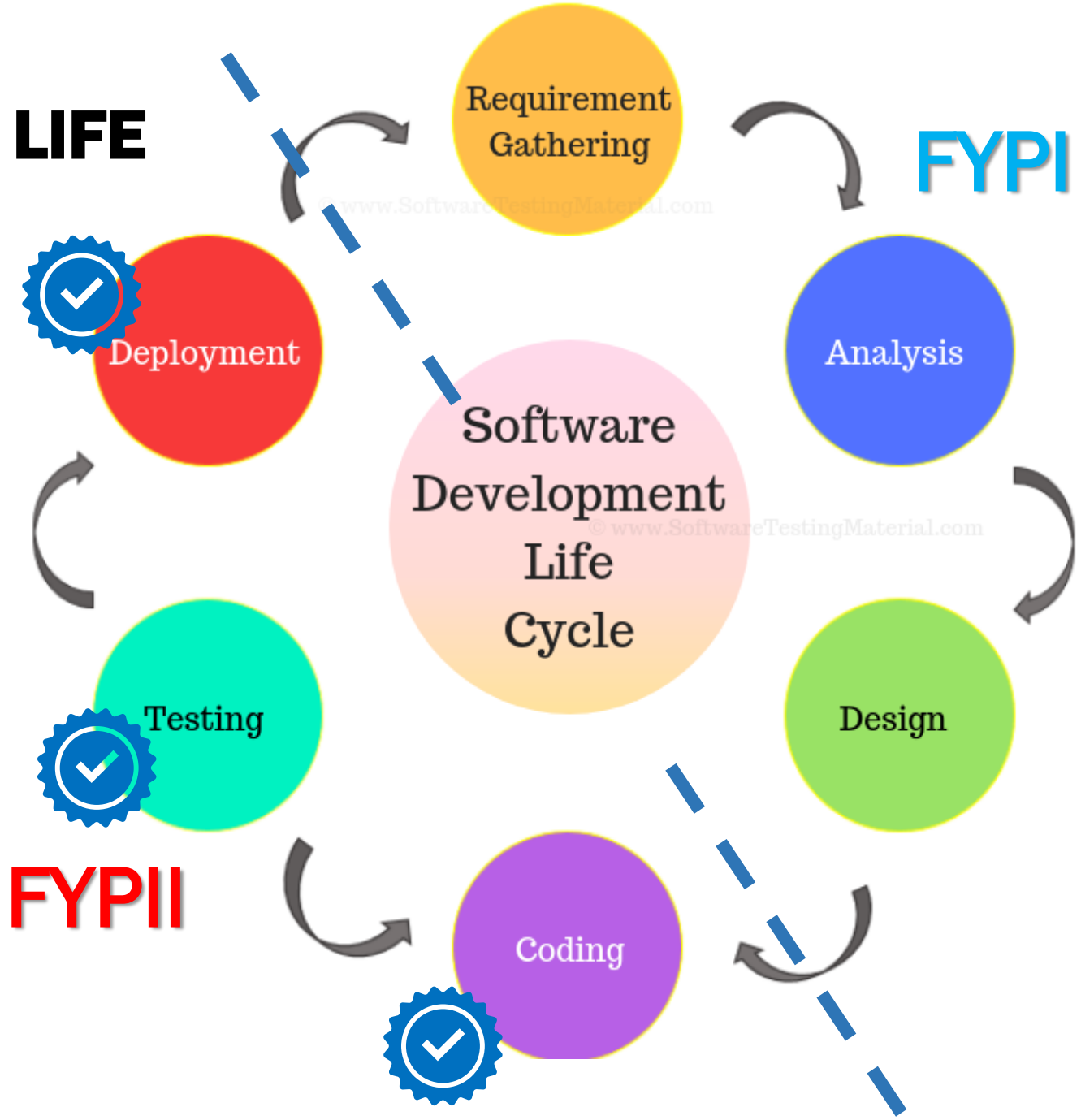
TS DR NUR RAZIA MOHD SURADI
COMPUTING DEPARTMENT
SOFTWARE ENGINEERING CLUSTER,
FCVAC, UNISEL

BK5 - 20th JANUARY 2026

SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

- A process followed for a software project, within a software organization. It consists of a detailed plan describing how to develop, maintain, replace and alter or enhance specific software.

- To produce a high-quality system that meets or exceeds customer expectations.



1) SYSTEM IMPLEMENTATION / TESTING / INSTALLATION

a) System Implementation/Coding

- To convert physical design specifications are turned into working and reliable software
- To document work that has been done.
- To develop the software

b) Testing

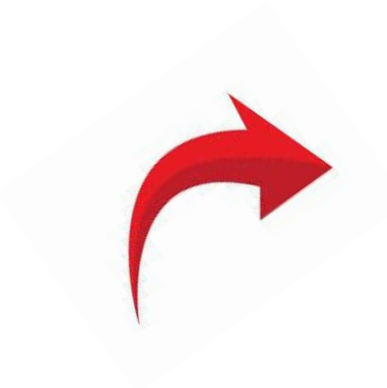
- Tests are performed using various strategies eg Dynamic or Static Technique
- Testing can be performed in parallel with coding

c) Installation

- Process during which the current system is replaced by the new system

FROM FYP1 - > FYP2

- CHAPTER 1 – INTRODUCTION ✓
- CHAPTER 2 – LITERATURE REVIEW ✓
- CHAPTER 3 – METHODOLOGY ✓
- CHAPTER 4 – ANALYSIS ✓
- CHAPTER 5 – DESIGN ✓
- CHAPTER 6 –SYSTEM IMPLEMENTATION AND TESTING
- CHAPTER 7- CONCLUSION



FUNCTIONAL REQUIREMENT
&
NON-FUNCTIONAL REQUIREMENT

1) SYSTEM IMPLEMENTATION

Coding

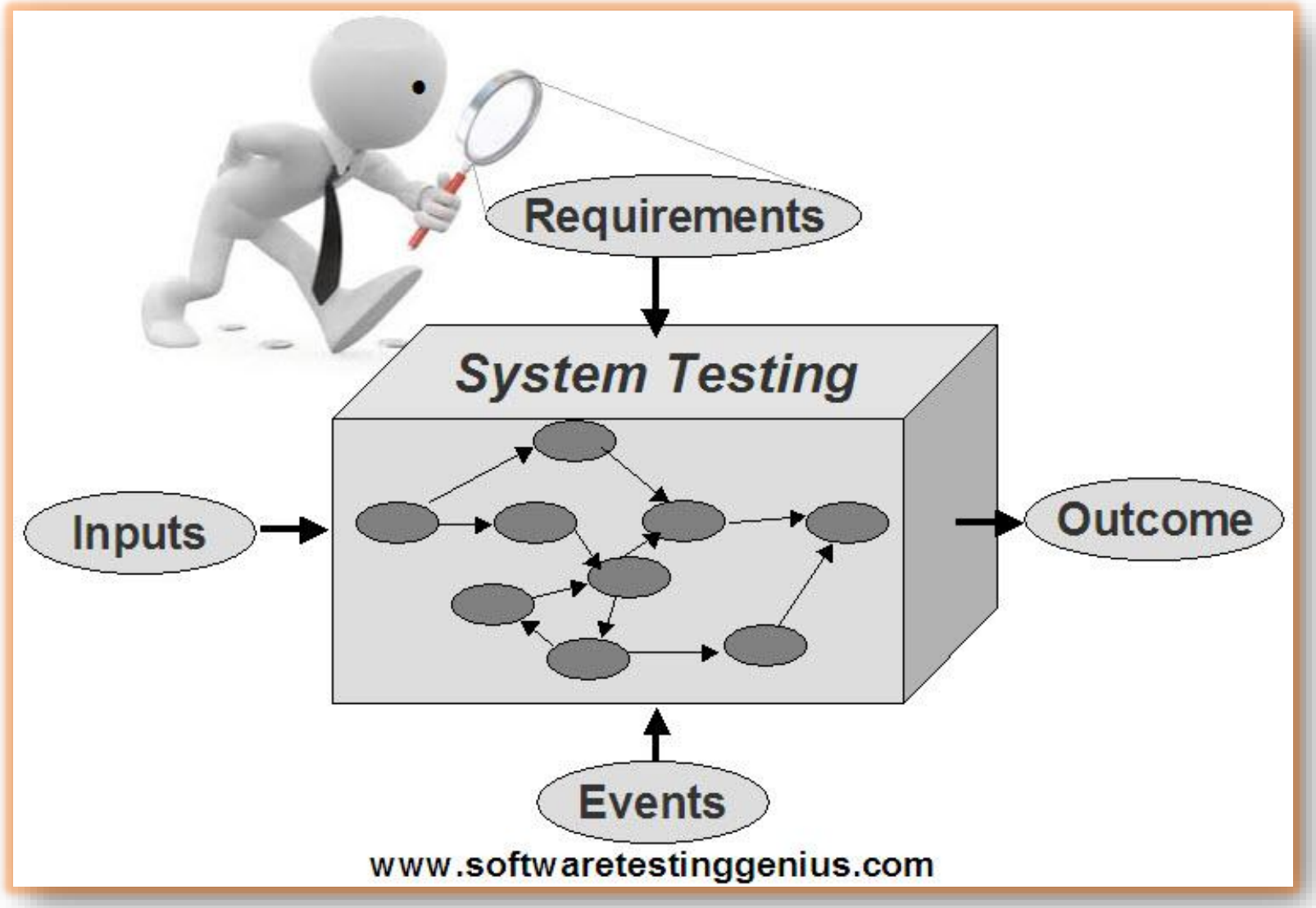
- Physical design specifications are turned into working computer code

Development Tools

- Programming language to used
- IDE used e.g. Android Studio, NetBeans, Microsoft Visual Studio etc
- Database GUI eg PhpMyAdmin etc

2) WHAT IS TESTING

- A process to **evaluate the functionality** of a software application with an **intent to find** whether the developed software met the **specified requirements** or not
- To **identify the defects** to ensure that the product is defect-free in order to produce a quality product.


Requirements
System Testing
Inputs
Outcome
Events
www.softwaretestinggenius.com

FUNCTIONAL TESTING AND NON-FUNCTIONAL TESTING

FUNCTIONAL TESTING	NON-FUNCTIONAL TESTING
▪ To verify that each function of the software application behaves as specified in the requirement document.	▪ How well the system performs is non-functionality testing.
▪ Testing all the functionalities by providing appropriate input to verify whether the actual output is matching the expected output or not.	▪ Non-functional testing refers to various aspects of the software such as performance, load, stress, scalability, security, compatibility, etc.,
	▪ The Main focus is to improve the user experience on how fast the system responds to a request.



OUR SCOPE – FUNCTIONAL TESTING

TYPES OF TESTING

Unit Testing:

- Also called as Module Testing or Component Testing.
- It is done to check whether the **individual unit or module** of the source code is working properly. It is done by the developers in developer's environment by **using sample input and observing its corresponding outputs.**

Integration Testing:

- Testing the fully **integrated modules** to evaluate the system's compliance with its specified requirements
- A typical software project consists of multiple software modules, coded by **different programmers.**
- Verifying the completed system to ensure that the application works as intended or not.

System Testing:

- **System Testing** is a level of testing that validates the complete and fully integrated software product.
- Performed after the integration testing and before the acceptance testing.

TYPES OF TESTING (CONT'D)

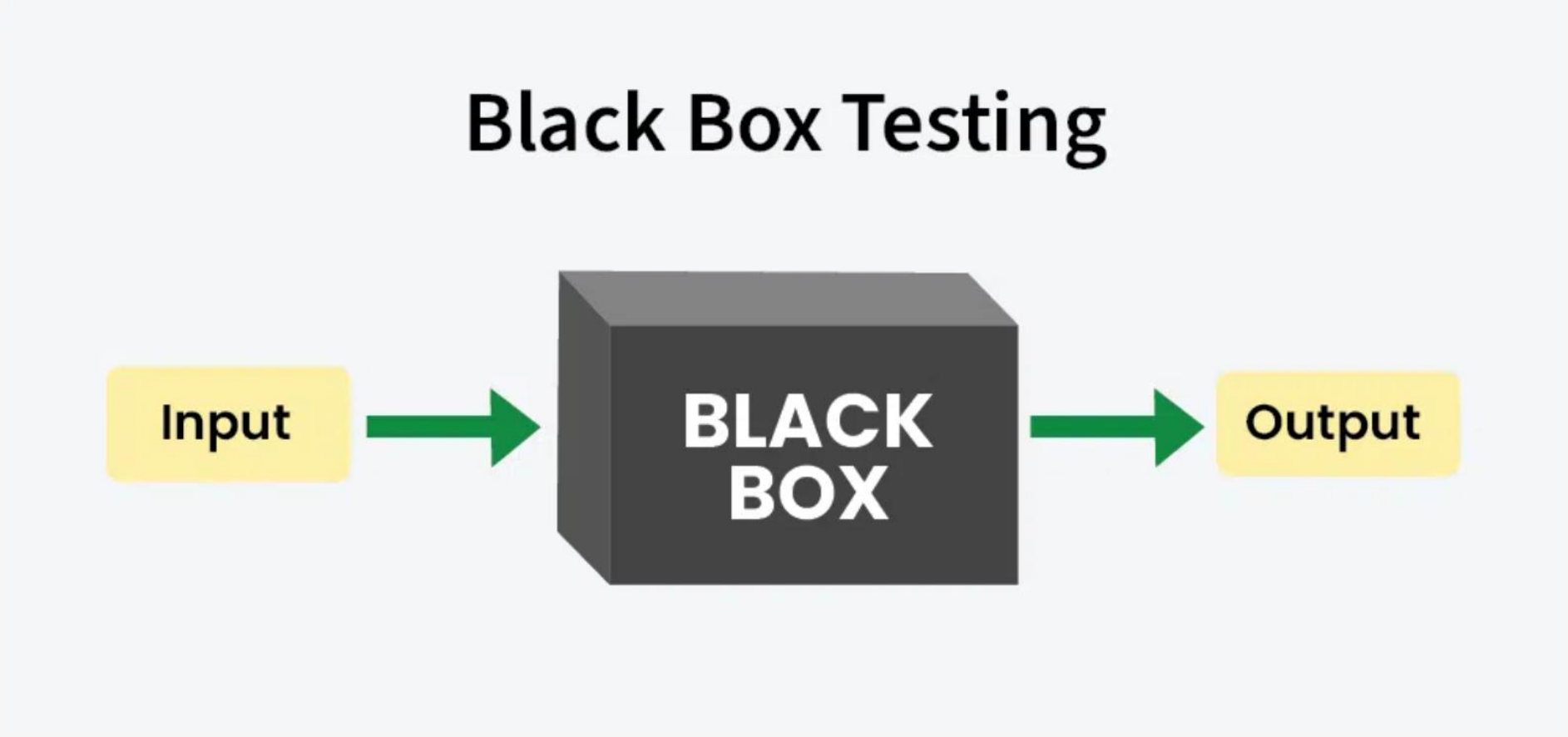
Acceptance Testing:

- It is also known as pre-production testing.
- It is NOT functionality test.
- This is done by the **end users** along with the testers to **validate the functionality of the application**
- Formal testing conducted to determine whether an application is developed as per the requirement. It allows **customer to accept or reject** the application.
- **Alpha, Beta & Gamma Testing:**



- 3 types of acceptance testing are Alpha, Beta & Gamma.
  - Alpha – development side (in-house). **Developer or QA AND customer** participate.
  - Beta – limited **end-user** participate. Outside the organization. Using a real or production server.
  - Gamma – conducted before final release of software. Only **customer** participate.
- It covers only the **Black-Box** testing process and hence the entire functionality of the product will be tested.

BLACK BOX TESTING



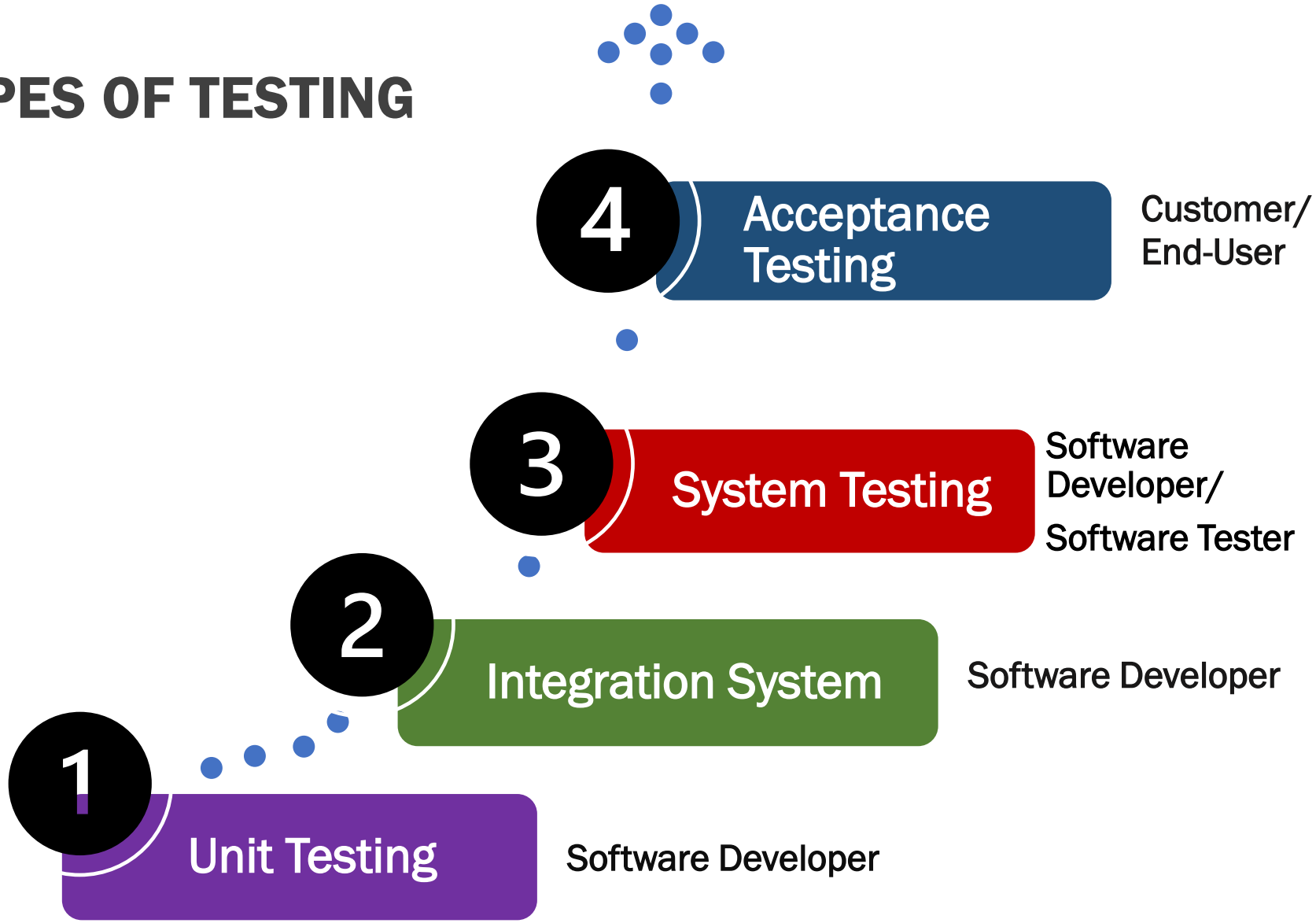
TYPES OF TESTING (CONT'D)

- In summary, there are 4 types of testing.



Unit Test	Test Individual component
Integration Test	Test Integrated component
System Test	Test the entire System
Acceptance Test	Test the final System

ROLE/TASK VS TYPES OF TESTING

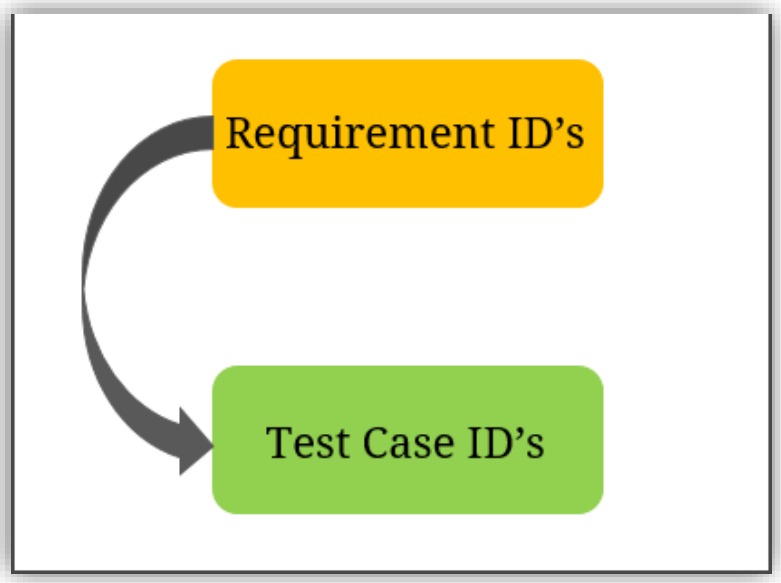


STEPS IN DOING TESTING

- Collect all the available requirement documents.
- Allocate a unique Requirement ID for each Requirement
- Create Test Cases for each requirement
- Link Test Case IDs to the respective Requirement ID.
- **REMEMBER:** Each Test Case ID **MUST MAP** with the Requirement ID.

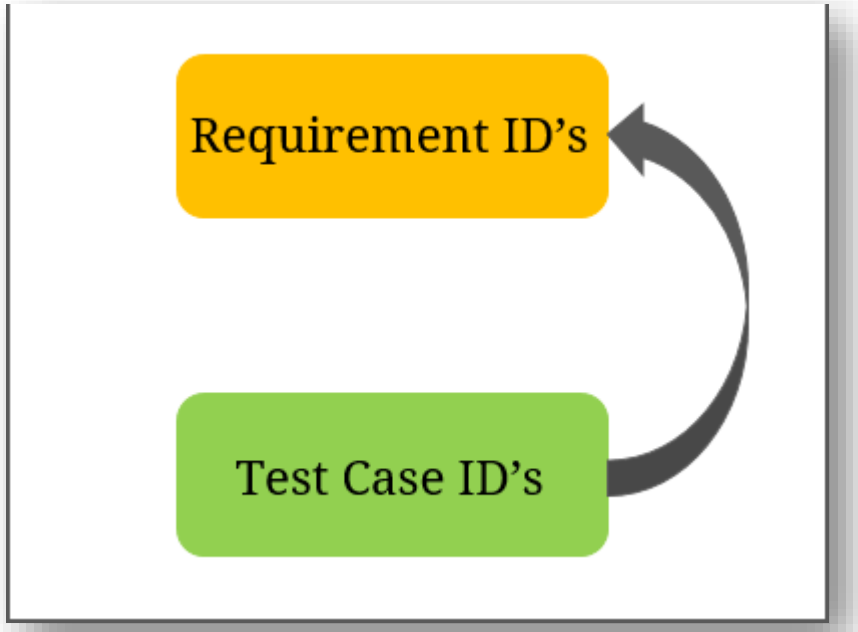
MAPPING REQUIREMENT-TEST CASE ID

1) Requirement ID -> Test Case ID



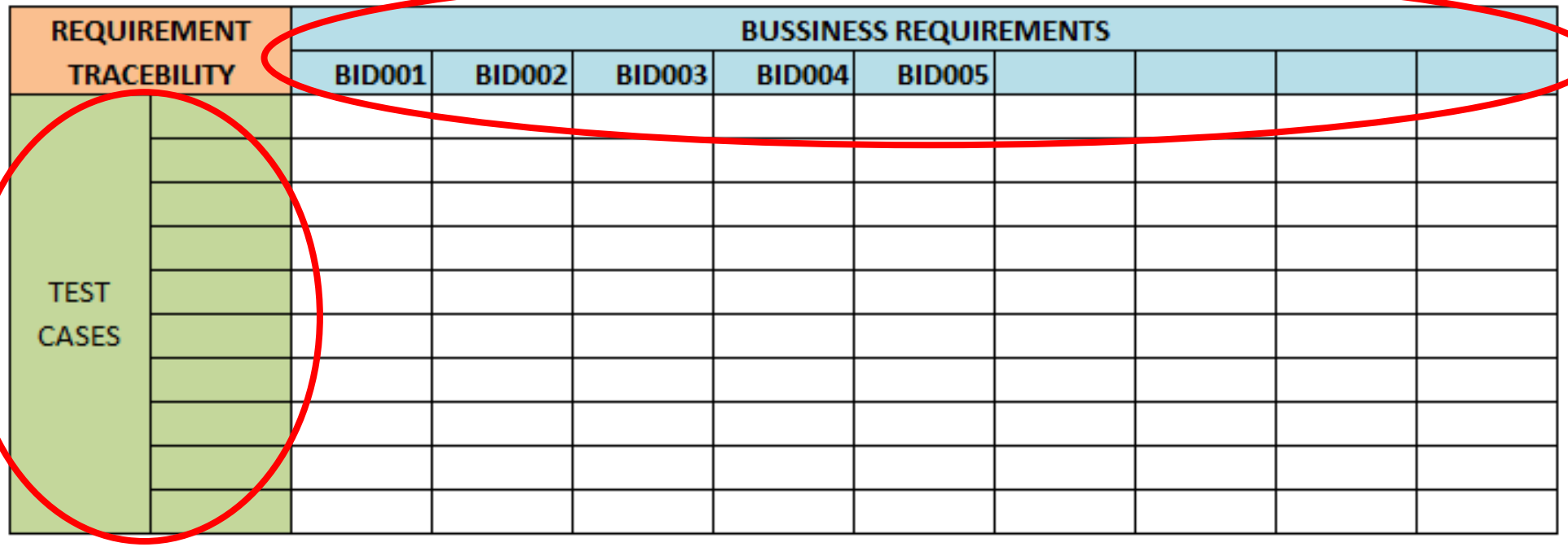
OR

2) Test Case ID ->Requirement ID



STEP 1 – CONSTRUCT BUSINESS REQUIREMENT (MODULE/ FEATURES) WITH TEST CASES TABLE

A group of business requirements eg BID001 Login Module, BID002 Reporting Module

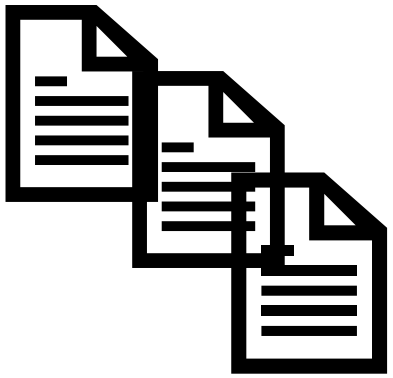


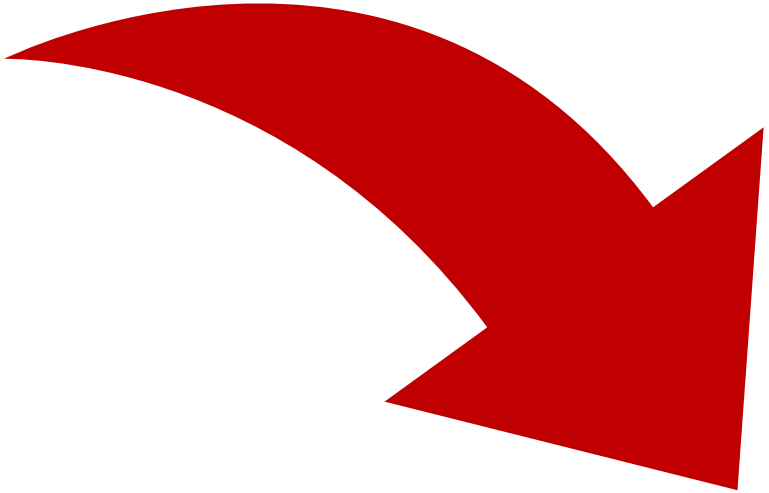
A group of test cases

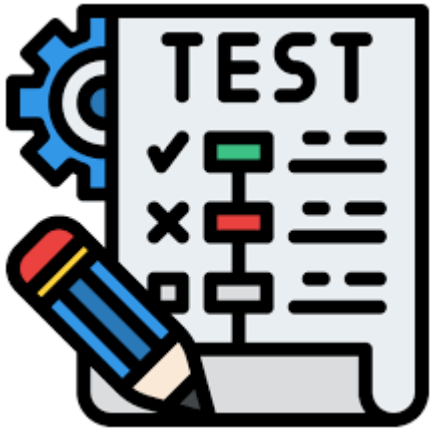
STEP 2 – MARK OR MAP EACH REQUIREMENT ID WITH TEST CASE ID

REQUIREMENT TRACEBILITY		BUSSINESS REQUIREMENTS								
		BID001	BID002	BID003	BID004	BID005				
TEST CASES	TID001	X								
	TID002	X								
	TID003	X								
	TID004		X							
	TID005		x							
	TID006			X						
	TID007				X					
	TID008				X	X				
	TID009					X				
	TID010					X				

FROM FYP1 DOCUMENT







TESTING MATERIALS


EXAMPLE

EXAMPLE :USE CASE DIAGRAM

- Use case diagram can be in one diagram OR separate by actors

- Actor - Admin and Patient
- Functional Requirement –
  - Login,
  - Manage User Profile ,
  - Manage Medical Report,
  - Manage Appointment
- Relationship -
  - extend and include



DETAILS OF USE CASE-TABLE OF DESCRIPTION

5.2.2.1 Detail of use case

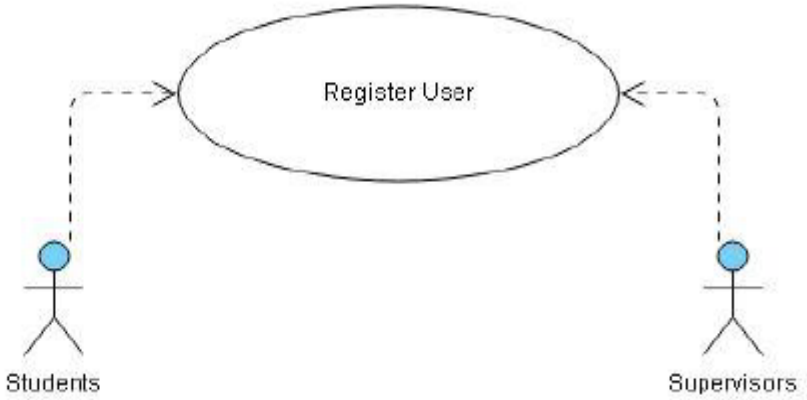


Figure 18: Use case of register user

Table 7: Descriptions of register user

Descriptions	
Use case name	Register user
Brief descriptions	The features that's allow supervisor and students that will practicum to register new user
Owner	Supervisors, Students
Actor	Supervisors, Students
Pre-condition	System display Sign Up Page
Basic flow	1. Insert Matric or Staff Id 2. For students, they will choose two option level education 3. Insert Full Name 4. Insert IC number 5. Insert Hp Number 6. Insert Email
Alternative flow	Not applicable
Post-condition	System display Log In Page

DETAILS OF USE CASE-TABLE OF DESCRIPTION

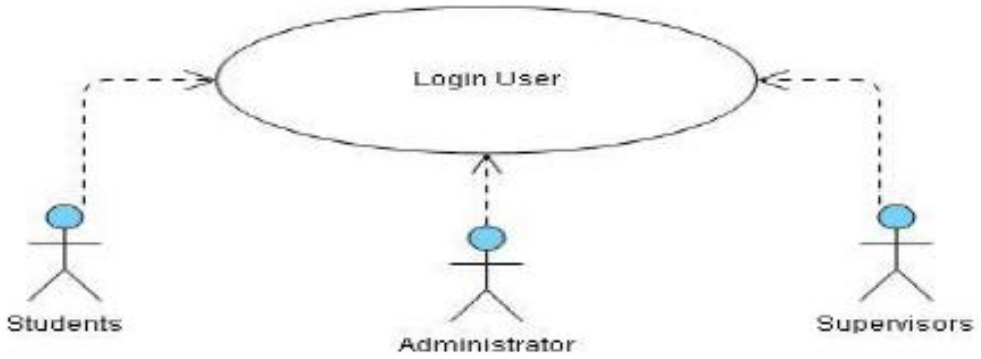


Figure 19: Use case of log in user

Table 8: Descriptions of log in user

Descriptions	
Use case name	Log in user
Brief descriptions	The users need to login in order to access their account. The users are requires to enter their username and password. After that the database validates the user to access into the application or system.
Owner	Admin, Supervisors, Students
Actor	Admin, Supervisors, Students
Pre-condition	System display login page
Basic flow	1. Insert matric number and passwords 2. Validate matric number and password from the database
Alternative flow	Re-login if matric number and password are wrong
Post-condition	1. For students and supervisors, after login they will go into view and edit their profile 2. For administrator, after login will go into monitor dashboard

Identify Functional Requirement from Use Case Diagram

1) Login
2) Register
3) Manage User Profile
4) Manage Medical Report
5) Manage Appointment

Functional Requirement



EXAMPLE LOGIN MODULE

Requirement ID	Requirement Description	Use case identifier and name	User
REQ001	Student login into the system by enter username and password	UC00 1	Student
REQ002	Admin login into the system by enter username and password		Admin

1) UNIT TESTING – LOGIN MODULE/ UC001

Requirement ID	Test Case ID	Test Case Description	Test Data	Expected Result	Actual Result	Date and Time Tested	Pass (P)	Fail (F)
REQ001	TC001	Student login into the system by enter correct username and password	• Username - string • Password - string	Login successfully. Redirect to Home screen.	Student login into system by enter correct username and password	12/5/2024, 2.30pm	P	
	TC002	Student login into the system by enter wrong username and password	• Username - 123 • password - 123	Error message display. Ask the user to enter correct username and password	Student login into system by enter wrong username and password	12/5/2024, 2.30pm	P	
						TOTAL (PASS/FAIL)	2	

2) INTEGRATION TEST - MANAGE APPOINTMENT/ UC002

Requirement ID	Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Date and Time Tested	Priority	Pass (P)	Fail (F)
REQ002	TC001	Create valid appointment	1. Open appointment form 2. Fill all required fields 3. Click Save	Date: 2025-06-15Time: 14:00User ID: 123Desc: “Meeting”	Appointment created successfully		12/5/2024, 2.30pm	High	P	
	TC002	Missing required field	Submit form without date	Time: 14:00User ID: 123Description: “Call”	Error: “Date is required”		12/5/2024, 2.30pm	High	P	
	TC003	Date in the past	Fill form with past date	Date: 2024-01-01Time: 09:00User ID: 123	Error: “Cannot schedule appointment in the past”		12/5/2024, 2.30pm	High	P	
	TC004	Overlapping appointments	Try to create an appointment at same time as existing one	Existing: 14:00–15:00New: 14:30–15:30User ID: 123	Error: “Appointment overlaps with existing one”		12/5/2024, 2.30pm	Medium	P	
	TC005	Invalid date format	Enter invalid date format	Date: “32/13/2025”	Error: “Invalid date format”		12/5/2024, 2.30pm	Medium	P	
	TC006	Invalid time format	Enter invalid time	Time: “25:61”	Error: “Invalid time format”		12/5/2024, 2.30pm	Medium	P	
	TC012	Back-to-back appointments	Schedule two appointments at 14:00 and 14:30	14:00–14:30 and 14:30–15:00	Both appointments created successfully		12/5/2024, 2.30pm	Medium	P	
	TC013	Appointment minimum lead time	Schedule appointment 1 minute from now	Current time: 10:00Appointment: 10:01	Accept/reject based on system policy		12/5/2024, 2.30pm	Medium	P	
									TOTAL (PASS/FAIL)	**8**

3) USER ACCEPTANCE TEST (UAT)

Table 6.1 UAT 1 - Login a User

Purpose	Precondition	Test Input	Steps to be Execute
Login User with an email and password	User has already registered	User fills up the required details	1. Tap 'sign in' text 2. Enter a valid email 3. Enter a valid password 4. Tap 'sign in' button
Expected Output	User is able to log in and is redirected to the Home UI		
Pass/Fail	Pass		
Date Completed	2nd February 2021		
Test Case Outcome	Satisfactory/ Pending for Clarification/ Return to Developer		
UAT Completed By	 Elissa Ahmad, Manager ABC Sdn Bhd		

3) USER ACCEPTANCE TEST (UAT)

Table 6.2 UAT 2 - Register a User

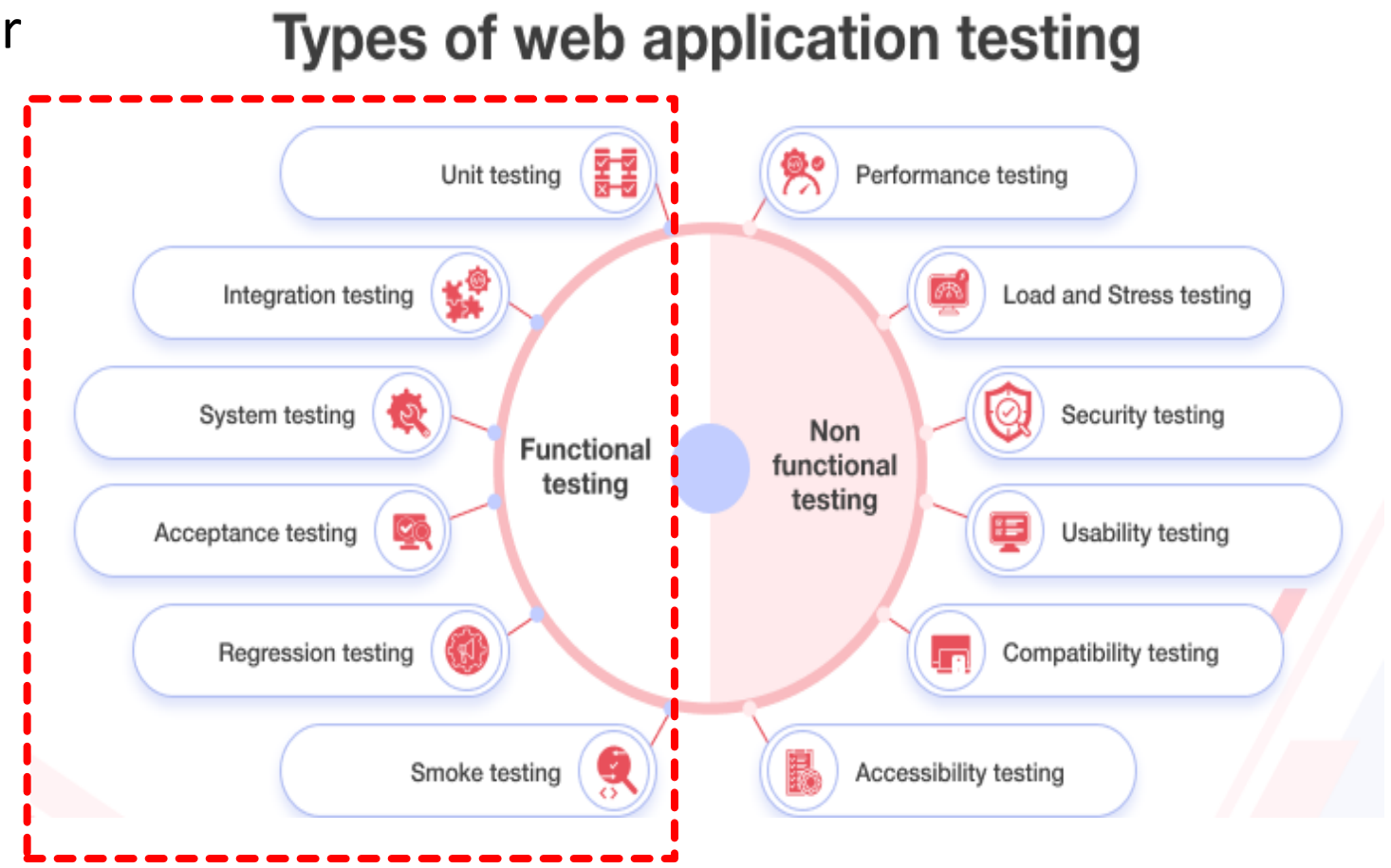
Purpose	Precondition	Test Input	Steps to be Execute
Create a user account	User has the downloaded the mobile application	User fills up the required details	1. Enter a valid email address 2. Tap 'continue' button 3. Enter a first name 4. Enter a last name 5. Tap 'next' button 6. Enter a valid phone number 7. Tap 'sign up' button
Expected Output	User is able to register and is redirected to the Profile UI		
Pass/Fail	Pass		
Date Completed	2nd February 2021		
Test Case Outcome	Satisfactory/ Pending for Clarification/ Return to Developer		
UAT Completed By	 Elissa Ahmad , Manager ABC Sdn Bhd		

TYPE OF APPLICATION

- WEB-BASED APPLICATION
- MOBILE APPLICATION
- IOT
- AI
- MULTIMEDIA APPLICATION

A) WEB APPLICATION TESTING

- Testing on development server (local host) **AND**
- Testing on real server (production/online)



B) MOBILE APPLICATION TESTING

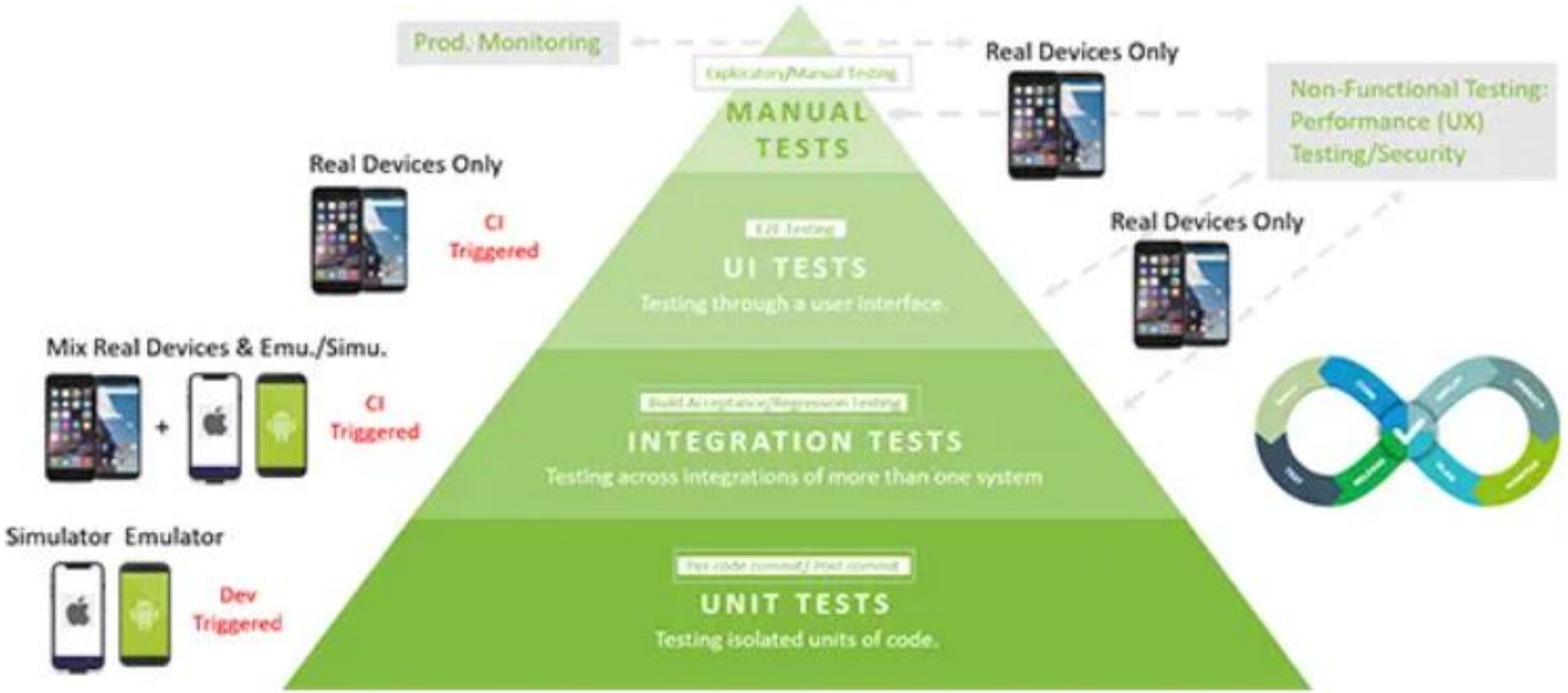
- Mobile application testing is very different from **software testing** and **web testing.**
- There are a few specifics that you should consider before performing mobile application testing:
  - Screen resolution
  - Turning on/off GPS
  - Screen orientation (landscape, portrait)
  - Different devices' manufacturers
  - OSs
  - Type of mobile application



MOBILE APPLICATION TESTING

Types of Mobile Application

- The type of mobile app you are testing plays a very important role in defining your testing process. So, be very clear about which type of mobile app you are going to test.
  - **Mobile web applications**: These are the web pages that you open through the mobile browser. **(HTML5, CSS, JavaScript, Ruby, and similar programming languages used for web work).**
  - **Native Apps**: These are the applications that are developed for **ONE** particular platform. (iOS, Android, Windows 10 M0bile, Tizen, BlackBerry). **(Some examples include: Java, Kotlin, Python, Swift, Objective-C, C++, and React.)**
  - **Hybrid**: It is combination of mobile web app and Native App. **(use a mixture of web technologies and native APIs. They're developed using: Ionic, Objective C, Swift, HTML5, and others.)**


Prod. Monitoring
Real Devices Only
MANUAL TESTS
Non-Functional Testing: Performance (UX) Testing/Security
Real Devices Only
CI Triggered
UI TESTS
Testing through a user interface.
Real Devices Only
Mix Real Devices & Emu./Simu.
CI Triggered
INTEGRATION TESTS
Testing across integrations of more than one system
Simulator Emulator
Dev Triggered
UNIT TESTS
Testing isolated units of code.

KEY FACTORS INFLUENCING MOBILE APP TESTING

- **Mobile Devices:** Mobile application testing follows a device testing approach - necessary to test our application in various resolutions, browsers, mobile devices, and operating systems.
- **Mobile Simulators:** Mobile simulators play an essential role in mobile testing. A simulator is a software that simulates the behavior of an actual device, or we can say it mimics the behavior of an actual device. These simulators provide a cost-effective way to test on various mobile devices without buying actual devices.
- **Network Conditions:** It is essential to test our mobile applications under various network conditions to ensure that they behave perfectly under networks like 2G,3G, or 4G.

C) IOT

What is IoT?



- The Internet of Things is about **connecting not just computers/phones but a whole range of objects and spaces to the internet**, enabling remote data gathering and interaction
- An IoT system has the following parts −
  - Sensor
  - Application
  - Network
  - Backend; also known as data center
- IoT is **a connection of embedded devices along with the internet infrastructure**.

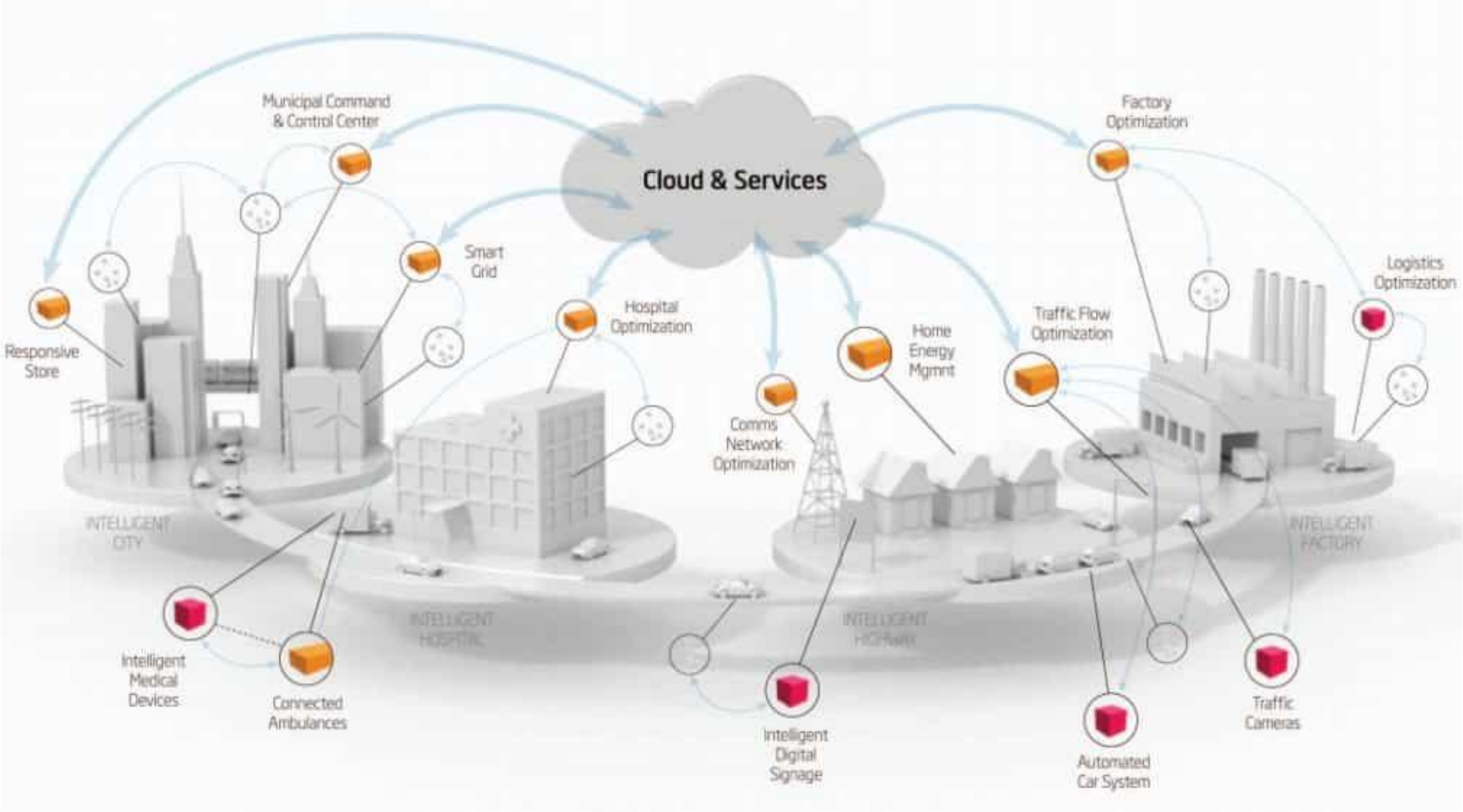
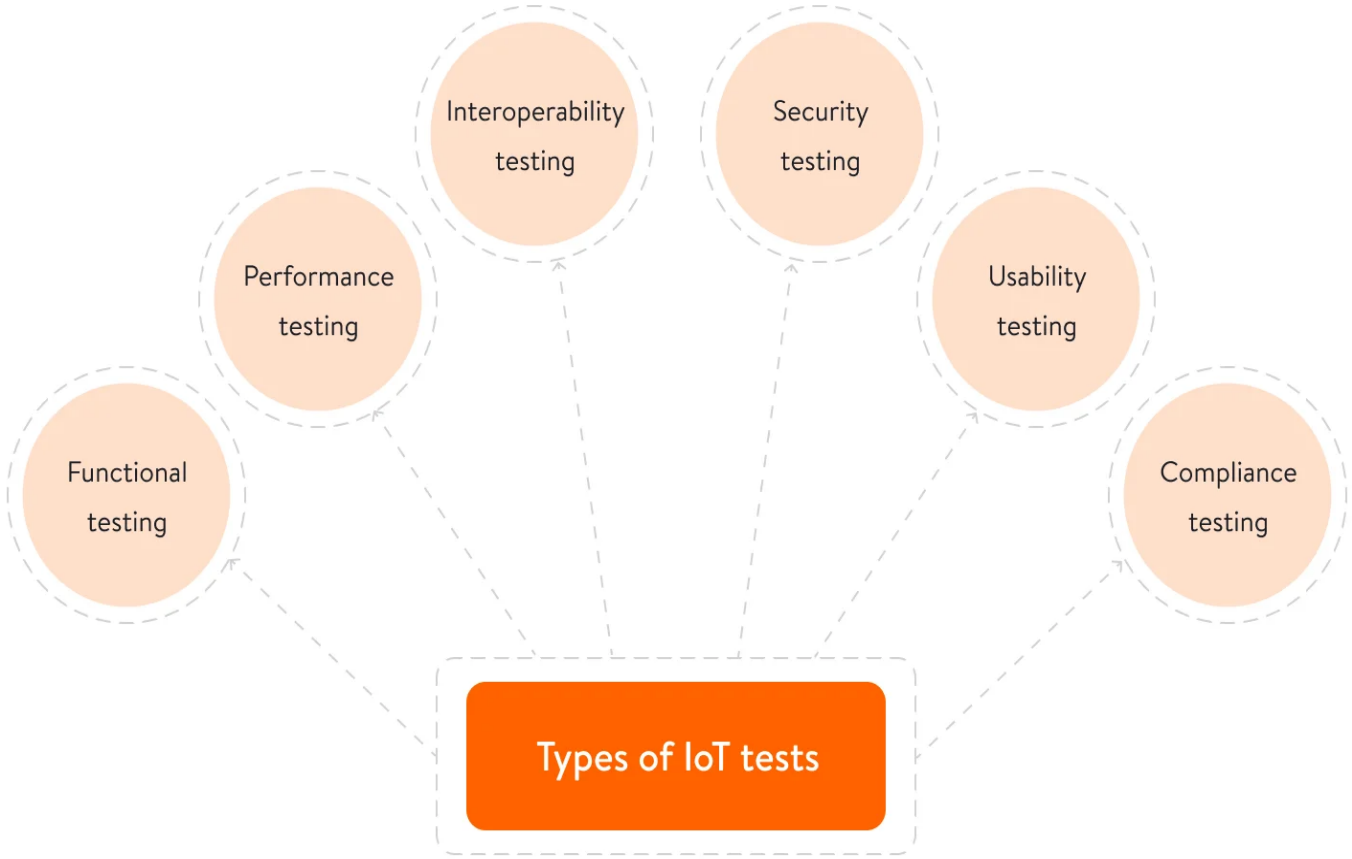


Image Credit: Opinno

TYPE OF IOT TESTING



- **Functional testing** ensures that the IoT solution works as intended by testing device interactions and user interface (UI) performance.
- **Performance testing** checks responsiveness, stability, and speed under typical and peak loads.
- **Interoperability and compatibility testing** ensures smooth communication between different devices, platforms, and protocols.
- **Security testing** identifies vulnerabilities, weaknesses, and threats through data encryption and privacy testing, physical security testing, and other methods.
- **Usability testing** evaluates real user interactions to enhance the user experience (UX).
- **Compliance testing** validates adherence to industry standards and government regulations.
- **Network testing** checks communications protocols like MQTT, Zigbee, and LoRaWAN.
- **Resilience testing** verifies system reliability under adverse conditions like hardware failures.

- Source: https://yalantis.com/blog/iot-testing-guide/

Functional testing include testing for **the data coming from things, environments, and other devices, commands coming from the cloud app, user actions spanning across the applications on the cloud and the device, device management triggers coming from the cloud applications**

IOT TESTING

Prerequisites of IoT Testing

- ✓ **Setting up IoT device** – The IoT device must be turned on, and can be accessed and used in real life. **E.g., while testing a smart watch, make sure to wear it on wrist.** Placing it on a table would not be regarded as a real user case.

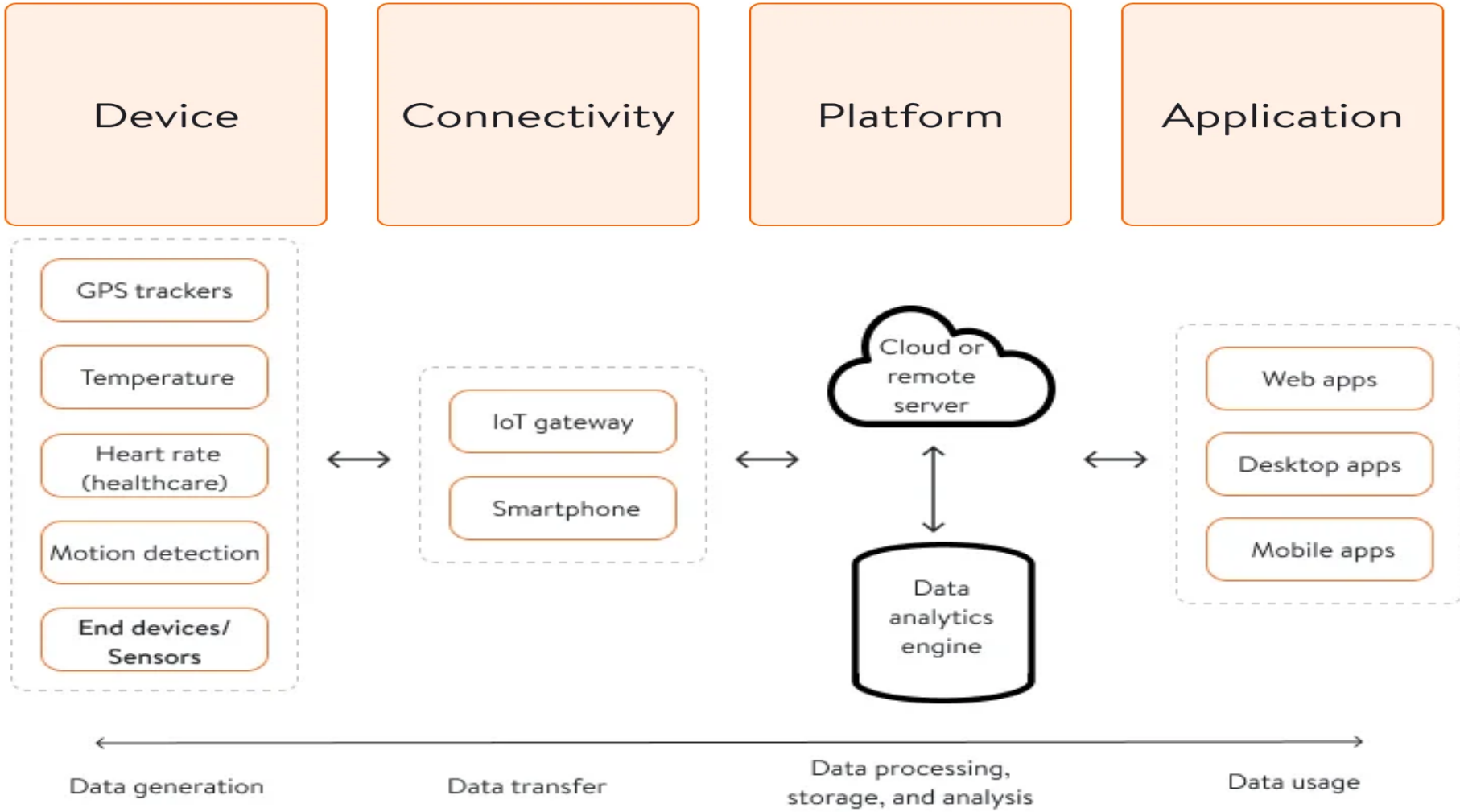
- ✓ **Setting up of IoT Hub** – IoT hub is a server that can connect with IoT devices and gather information from them. An IoT hub may be **an application in a mobile device or a web server on a cloud.** The IoT hub must be set up properly.

- ✓ **Setting up network** – A strong **wireless connection to connect IoT hub and the IoT device** together. This can be possible with a Wi-Fi, Bluetooth, satellite signals, NFC (near field communication), etc.

*** While connecting wearable device with a mobile app, ensure the following –

- The Bluetooth of both the devices is turned on.
- Both the devices are paired together.
- Both the devices are in range of each other.

COMPOSITION OF IOT SYSTEM



[Composition of an IoT system]

TYPES OF IOT TESTING FOR VARIOUS COMPONENTS OF IOT

IoT Testing Types	Sensor	Application	Network	Backend
Functional testing	True	True	False	False
Usability testing	True	True	False	False
Security testing	True	True	True	True
Performance testing	False	True	True	True
Compatibility testing	True	True	False	False
Services testing	False	True	True	True
Operational testing	True	True	True	True

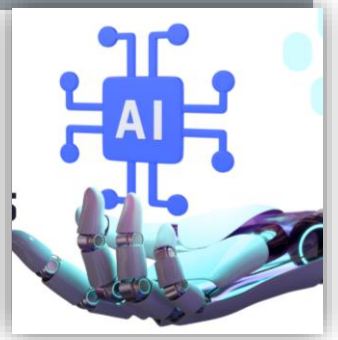
DATABASE ON IOT

IoT data is saved in TWO ways:

- Data storage on a local or cloud server.
- Data storage on the IoT devices' internal memory.

- The most common example where data is saved is would be on **the internal drive of your computer or phone.** Removable media such as USBs sticks or DVDs also constitute local storage.
- Example relational databases like PostgreSQL; there are also many IoT NoSQL databases like Cassandra and ScyllaDB

D) AI



- Traditional testing methods, such as test coverage, are often ineffective when testing machine learning applications
- The efficiency of AI systems is based on the **quality of training data**, including aspects like bias and variety AND **the evaluation of the AI models, algorithms, and machine learning (ML) systems** that drive the application AND **User Experience(UX)**

STEPS FOR AI TESTING



Data collection. AI models need data to learn from. This data could be anything from images, text, numbers, or even sound. For instance, if you want to build a model that can recognize cats in pictures, you will need a lot of pictures of cats.

Training.

Once you have your data, you use it to train your model. This is like teaching the model what a cat looks like. You show it lots picture of cats and tell it, “This is a cat.” The model learns from this data and starts to understand what features make up a cat.

Testing.

After training, you test your model to see how well it has learned. You show it new pictures of cats that it hasn’t see before and ask it tell you if there’s a cat in the picture. If it gets it right most of the time, you know you trained and tested, you can use make predictions on new data. For example, if new picture of a cat, it should be able to tell you, “Yes, that’s cat!.”

Prediction.

When your model is trained and tested, you can use it to make predictions on new data. For example, if you show it a picture o a cat, it should be a you. “Yes, that’s a cat!.”

STEPS FOR AI TESTING (CONT'D)



- **Usability and User Experience (UX)**

- **UI/UX Testing**: Test the user interface and experience, particularly in how AI-generated outputs (e.g., recommendations, predictions, classifications) are presented to the user.
- **Human-in-the-loop (HITL)**: Evaluate any user intervention steps in the process (e.g., users correcting AI mistakes or confirming predictions).
- **Transparency**: Assess if the AI's decision-making process is communicated clearly to the user, especially for critical applications like medical diagnosis or finance.

E) MULTIMEDIA



What is Multimedia application

- Is an application that manipulates **text, data, images, sound and full-motion-video objects**.
- Multimedia Testing involves testing your multimedia products on these different formats and mediums.

Multimedia application testing

- Performing multimedia application testing involves ensuring the application functions as expected across various devices, operating systems, media formats, and scenarios.

STEPS FOR MULTIMEDIA APPLICATION TESTING:



- **1. Define Testing Scope**
- Identify the type of multimedia application (e.g., video, audio, interactive media, games).
- List the features to be tested (e.g., media playback, recording, editing, streaming, transitions, user interface).
- **2. Set Up Testing Environment**
- Prepare different devices, operating systems, and browsers to test compatibility.
- Ensure you have various multimedia formats (audio, video, images) available for testing (e.g., MP4, MP3, JPG, PNG).
- Install relevant plugins, codecs, and software (e.g., Flash, QuickTime, Media Players).
- **3. Test Compatibility**
- **Cross-browser testing**: Test the application on different browsers (Chrome, Firefox, Safari, Edge).
- **Cross-device testing**: Test on a variety of devices (desktop, mobile, tablets).
- **OS compatibility**: Ensure functionality on Windows, macOS, Linux, iOS, and Android.

STEPS FOR MULTIMEDIA APPLICATION TESTING (CONT'D):



- **4. Functional Testing**
- **Media Playback**: Verify audio and video files play correctly.
  - Test play, pause, stop, rewind, fast-forward, and volume controls.
  - Ensure subtitle display and audio synchronization if applicable.
- **Media Quality**: Test media resolution (HD, 4K, etc.), aspect ratio, color accuracy, and sound quality.
- **Media Format Support**: Ensure support for various file formats (MP3, AVI, MP4, MKV, WAV).
- **User Interactions**: Test features such as drag-and-drop, zoom, play/pause buttons, volume controls, etc.
- **Streaming**: If applicable, test streaming functionalities (buffering, quality change, and uninterrupted playback).
- **Editing**: Verify any editing tools work (e.g., trimming, cropping, effects).
- **Content Search**: Test search and filtering functionality (if available).

3) PREPARING INSTALLATION



- Once system already tested and UAT success, we need to prepare the installation at customer side
- Means, converting the new system to replace existing or old system .
- Proper **training** to guide the user in using new system eg hands-on
- **User Manual** of the system (steps-by-steps to use the system)









Why we need to do testing?

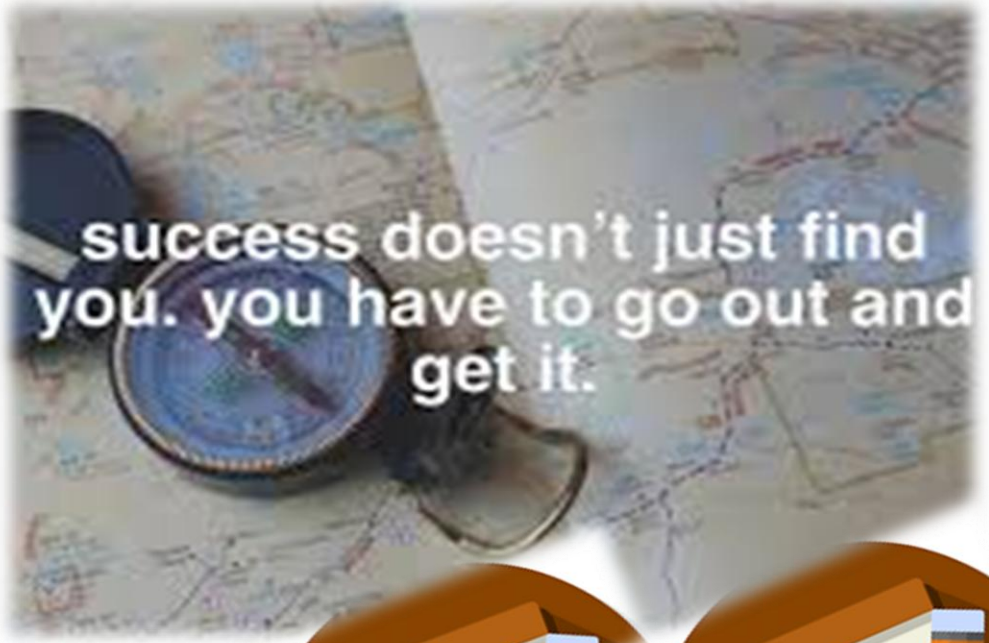


Why complete testing is impossible?

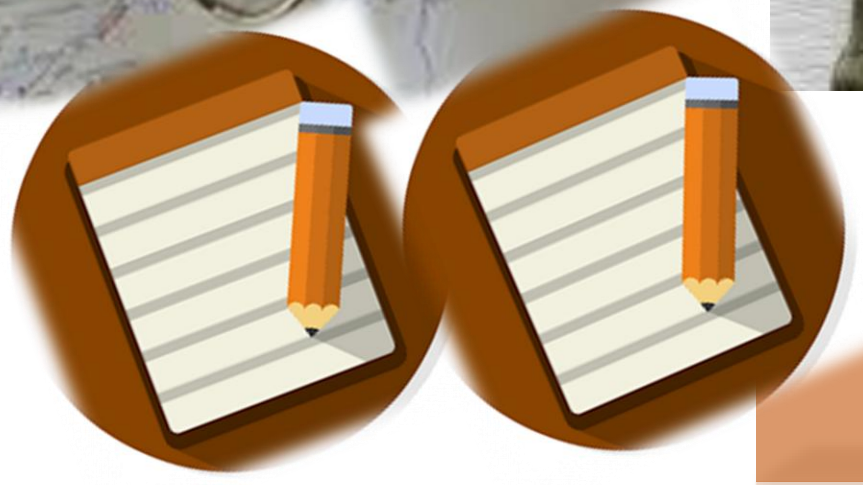


What are the possible testing types should we implement?

Q&A







GOOD LUCK & ALWAYS GIVE THE BEST